CHAPTER 8

LIFE PROCESSES: CONTROL AND COORDINATION



You have studied about several life processes. Let us consider a few of them.

While we swallow our food the process of breathing stops momentarily. During the process of digestion of food different types of substances are either secreted or absorbed in different parts of the alimentary canal (gut). As soon as something comes too close to our eyes, the eyelids are shut.

Regarding plants, we find that the leaves of the plant 'touch me not' (chuimui) close as we touch their tips. We often find the stomata open in several plants, during the process of photosynthesis. Such processes go on continuously.

Whether processes go continuously or a process stops as another goes on, there is a continuous exchange of information. A system for communication of information is present in the bodies of all living organisms that enable such functions to occur at certain times.

Let us study about some of these systems. We shall study about them first in humans. Conduction of information in our body is mainly carried out through a system of nerves and some chemicals.

8.1 Systems related to coordination and control in humans

Let us do an activity to observe how certain systems work in coordination with each other conducting information from one end to the other and how fast they do so.

Activity-1

Take a half meter scale or a stick. Keep your fingers around the stick as shown in figure1. Let a friend hold the stick at the top in such a manner that the lower end of the stick is suspended freely between your fingers (mark this point with a marker). Now let your friend release the stick, instructing you to hold it. Make an effort to hold it immediately.

- Could you hold it at the same point marked by you or is the point where you could hold the stick higher than the previous point?
- Why do you think this may have happened?



Figure-1: Stimulus (information) and its response

- How did you know that the stick fell?
- Which part of your body was responsible for holding the stick?
- What could be the other parts involved in the process? Enlist them.

8.1.1 Parts of body involved in conduction of information:

You may have observed in activity 1 that you had received information about the stick through your ears and eyes. You could hold the stick with your fingers. Other parts involved might have been the muscles of your hand, your brain, spinal cord, nerves etc.

The role of brain, spinal cord and nerves has been studied extensively from around 3000 years back. 'Shushrut' a medical practitioner during this time in India is believed to have known that brain had a very important role in exchange of information in our body. Techniques of cure by Shushrut are thought to have involved restoration of function in affected parts of the body and reduction of headache. Certain techniques of Shushrut are believed to be followed in Ayurveda even today. After around 500 years of Shushrut's time, Greek medical practitioners appreciated the role of spinal cord and nerves along with brain in the process of conduction of information. Nerves were believed to be agents of transfer of information and those that rendered movement to our body.

During this time, a Greek medical practitioner named 'Galen' made a remarkable observation while treating a patient. The patient had been injured near the neck and lost sensation in his hand. Yet the functional ability of the hand was not lost. Thus Galen concluded that there must be two pathways of flow of information in our bodies, one for sensations and the other for reactions. He was of the opinion that the brain played an important role in controlling both out sensations and reactions. Also, the spinal cord had a supportive role to the brain and was a medium of transfer of information.

Studies done during the 16th and 17th centuries suggested that the spinal cord not only aided functions of the brain but certain sensations and reactions were controlled by it. It was observed that even in the absence of the brain, sensation and movement in the muscles of the body could be observed for some time.

Further on, after the discovery of cells and observation of arrangement of nerve cells in the body, we also came to know about their probable structure, composition and function. Camillio Golgi who had discovered the Golgi body and Schwann who was one of the proponents of cell theory had independently studied the nerve cells and their distribution in our body. Golgi bodies were observed first in the nerve cells.

You have already observed the figure of nerve cells and tissues in class 9. Let us study the structure and function we know so far of nerve cells.

8.1.2 Structure of nerve cells:

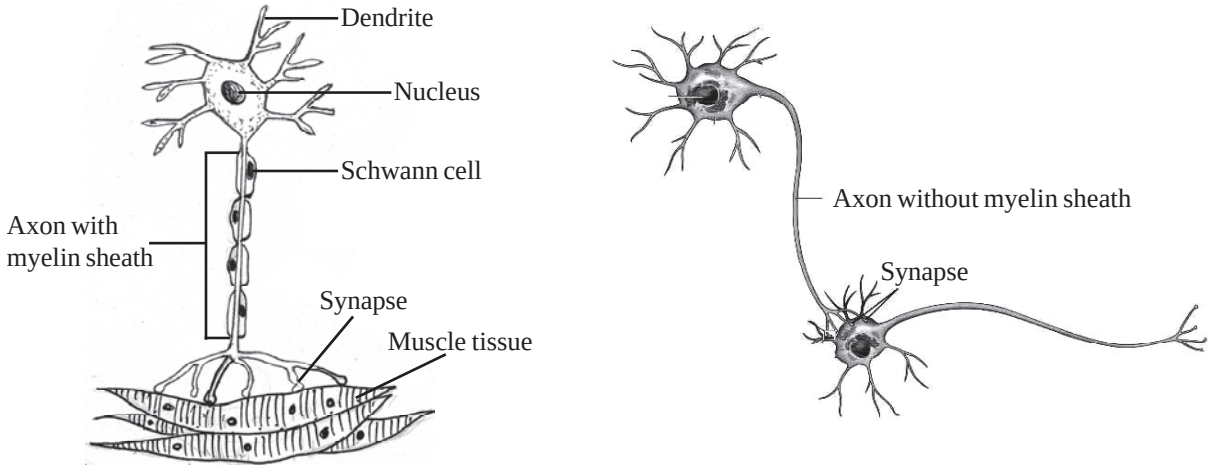


Figure-2: structure of nerve cells (A) with myelin sheath (B) without myelin sheath

Schwann had observed that nerves that carried sensations had fatty covering over them. After about two decades of this observation, a description of nerve cells was presented. This remained unaccepted to the scientific community for a long time as it was a challenge to observe these cells separately. On the one hand these are observed as mesh like structures while on the other it is difficult to separate them from muscle tissues.

After the advancement in staining techniques and microscopes, in the decade of 1910 these cells were observed clearly for the first time and their description given previously was now confirmed. Researches related to the arrangement, growth and development of these cells started during the 1930's. Thus it became clear that nerves were of different types and usually formed of groups of nerve cells. Each nerve cell had a nucleus and several mitochondria, Golgi bodies and other cell organelles.

The speciality of nerve cells is that their cell membrane and cytoplasm form extended branch like projections called as dendrites (dendrite: branched tree like). Some of these that are long are called as axons (axon: axis). Axons reach the furthest extremities of our body. Some of the axons are covered with a fatty layer called as myelin sheath. Schwann had observed that the sheath was made of cells which were later called as Schwann cells.

Do you know?

Camillio Golgi and Ramon Cajal were conferred the Nobel Prize in1906 for their description of the nerve cell. It was similar to the description given by Robert Remak way back in 1850's. Remak's description was unaccepted during his time.

The Schwann cells in a myelin sheath are wrapped up in such a manner over the axons that spaces remain between them. Myelin sheath renders protection and support to the axon and helps in

bringing together several axons in a bunch. It also helps in increasing the rate of flow of information. The axons of nerves with their cell bodies in the brain and spinal cords reach different areas of our body. You may be surprised to know that axons may be so long as to start from brain or spinal cord and reach the end of our toes. Thus some of our nerve cells are as long as our height!

- Where do you think is the cell body carrying nucleus of such nerve cells?
- Nerves have protective coverings, what are the protective coverings of our brain and spinal cord?

Apart from nerve cells there are several blood vessels and other mainly protective cells in our brain and spinal cord. All these structures are together wrapped by dermal tissues that have fluid filled between the layers of them. These are further covered by the bony armour of skull and vertebral column. Vertebrae have mainly shock absorbing fluid filled discs between them that also help in smooth movement.

Information is conducted between nerves or nerves and muscles through areas called as 'synapses'. Though the name synapse refers to fixed overlapping connections, in reality these have been observed to be areas of gaps either between nerves or nerve and muscles. Thus certain chemicals in these gaps help in conduction of information. So far we have come to know that the conduction of information across nerves is due to certain electrochemical processes. These are very complex and are a field of ongoing research.

8.1.3 Nerves and Sense organs

We are able to sense several types of sensations or stimuli like heat, sound, taste, light, pressure, smell etc.

- What are the different organs of our body that we may associate with these sensations?

Apart from the sense organs like the eyes, skin, nose, tongue, ears that we have studied earlier, muscles in areas deep inside our body also have areas that have sensations (think of the burning sensation in the stomach). Moreover a particular part may sense different stimuli like the skin may help to sense changes in temperature, touch, pressure, pain etc. All the areas of sensation have several nerve endings or nerve cells that help in the process of sensation. The axon terminals of these may synapse directly with dendrite or axon terminals of those of the brain or spinal cord or with certain connector nerves.

As for example, there are light sensitive nerve cells in the retina of our eyes. The axons of these synapse with the dendrites of certain nerve cells whose axons together pass to the brain(as nerve fiber passing through the retina: area called as blind spot). There they form synapses with the nerve cells in the brain (see figure 3). The axon terminals of nerves coming from the brain pick up sensations inside our nose. Due to such arrangements in our body and others, several functions go on in a coordinated manner in our body. A simple example would be - we often enjoy the taste of food when we get its flavor as well.

Activity-2

Eat a pinch of fennel seeds (saunf). Now wash your mouth well and close your nose and eat a pinch of fennel seeds again.

- Is the taste same in both situations?
- What may have been the cause of this?

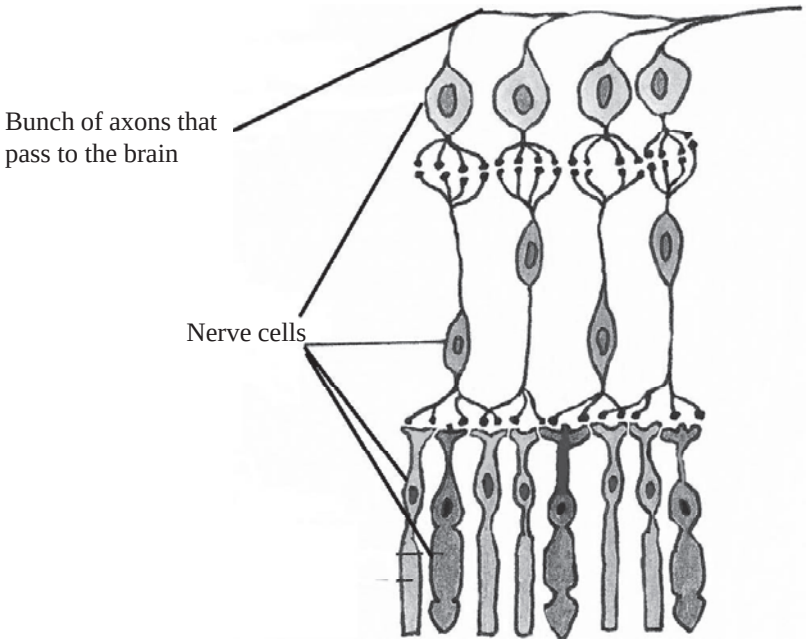


Figure-3:

8.1.4 Structure and function of spinal cord

We studied about the structure of nerves, their arrangement and functions in our body. We shall now study their arrangement in our brain and spinal cord. Spinal cord is a bunch of thread like structure running from brain through our vertebral column. It is composed of nerves. The myelinated axons of these nerves pass out through gaps between vertebrae and reach various parts of our body (represented by the white portion in the section of fig.4). The axons without myelin cover are located towards the inner side(represented by grey portion in the section of fig.4).

If a person's spinal cord somewhere in the middle of the back is injured such that no information passes to the brain, then there is a loss of sensation in the area of the stomach and that of legs. The person fails to move these parts on her own. But if the sole of the foot is tickled, the foot moves away without her knowing what may have happened!

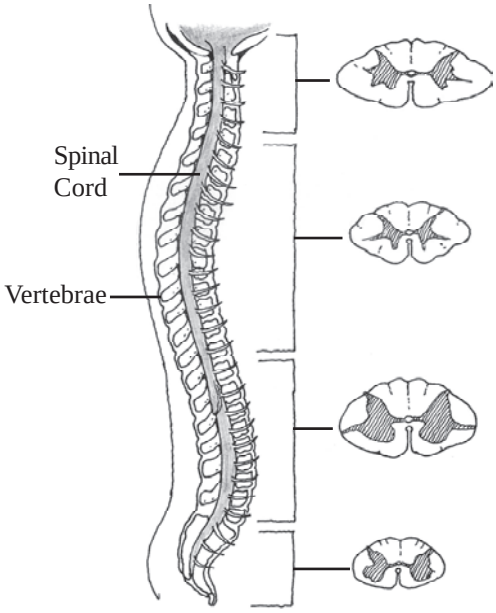


Figure-4: Spinal cord and transverse sections of its different areas

Several functions occur in our body without our conscious sensation of the same. For example digestion of food, flow of blood through various parts, secretion of digestive juices, movement of the hand as it touches a hot surface etc. We call such functions as involuntary functions as they go on without our conscious effort. Most of these are controlled directly by nerves of our spinal cord. The information of these of course passes to the brain.

8.1.5 Structure and Function of Brain

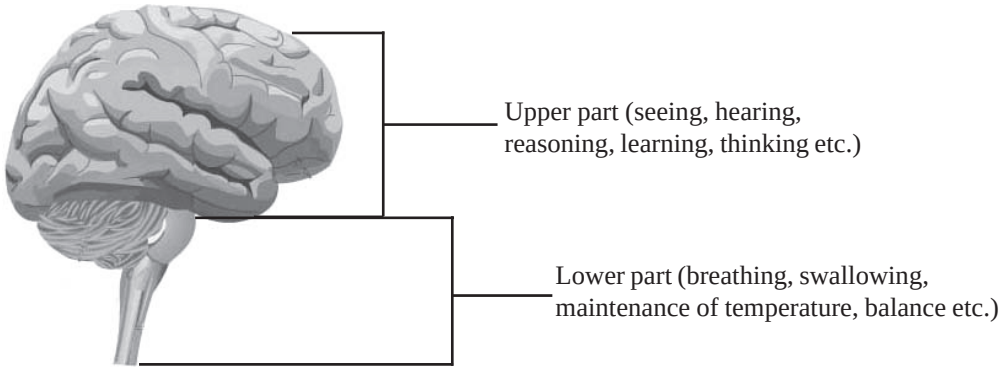


Figure-5: Human Brain

Brain as we know has a very important role in control and coordination of our body. The structure of the brain is very complex and it is very difficult to study the various functions that are carried out in our brain in detail. Scientists study certain sections of brain by arresting function of others to find out about the role of these specific sections. To know about structure and function of our brain we often try to study our brain in 3 parts- front, middle and back. The functions of front and middle as well as middle and back are so overlapping that it creates a lot of confusion to study the brain as such. Thus, it would be easier to state two parts of the brain as upper and lower parts. The upper having roles in seeing, hearing, smelling, reasoning, learning, thinking etc. while the lower part in breathing, swallowing, maintenance of temperature, balance etc. The upper part of our brain is most developed as compared to other animals of our family.

The brain also controls involuntary actions controlled by the spinal cord. As for example when you walk, your hands move in a particular manner according to your footsteps and this happens without your conscious sense about the action. This is when the spinal cord exclusively controls your actions. Contrary to this when you are rehearsing a march past; you move your hands and legs in a pattern intentionally in a conscious manner. This action is voluntary which is being controlled by the brain. Thus we may say that whenever we do something consciously or voluntarily, it's our brain that controls such actions. There are certain involuntary actions as well that are directly controlled by our brain like the contraction of pupil of our eye when exposed to bright light.

Do you know?

Our heart beat is stimulated and controlled by a network of nerves in our heart. It receives signals from the brain for it. But even if the signal stops from our brain the nerves may still keep the heart working if signaled artificially.

- Make a list of all the voluntary actions that you do in a day.
- Write the organs involved in any one of these voluntary actions.

Any of the following three functions (or all of them) may go on if the upper part of our brain functions normally.

1. It may arrest those functions that were stimulated by the lower part or the spinal cord.
2. It can carry the functions stimulated by lower part or spinal cord in a controlled manner.
3. Can control function of muscles directly without the intervention of spinal cord.

Well there are still several other amazing facts about our brain. Another fact is most of the left part of our brain controls functions of most of the right part of our body and vice versa. The responses to informations that reach the brain are expressed in a consolidated manner.

Let's do some activities to observe this.

Activity-3

Keep your hands in a manner as shown in the figure. Let the pointer fingers pint each other.

Now move both your hands in a clockwise direction together. Now roll one of them clockwise and the other anticlockwise simultaneously. Try stopping one to roll the other in clockwise and anticlockwise directions respectively.



Figure-6: Bringing pointer fingers in front of each other

- Was it easy to roll your handssimultaneously in opposite directions?
- Why do you think it was so?
- Why was it easy to stop one hand and move the other?

While you were rolling both hands together the response from the brain was a combined one. Directions for individual activities came separately. But when we tried doing antagonistic actions here, the combined response from the brain was directed otherwise.

- While walking which hand swings forward as you put your right leg out?

How do you think it happens like this?

Activity-4

Blindfold one student sitting in the center of the class. Ensure that her ears and nostril are not covered. Ask the student to respond to sound of clapping by pointing towards the direction. Now, the teacher may choose students sitting at equal distance behind and in front of as well as towards left and right side of the blindfolded student. These students will clap in turns as the teacher points towards them one by one. The blindfolded student will point towards the source of sound. Rest of the students must sit quietly and watch.

Repeat this with other students to find out whether the observation is similar with them.

- Could the blindfolded student always point correctly towards direction of clapping?
- What happened when the sound of clapping was either from front or from back of the student?

- Through which organ was the student getting the information of the sound stimulus?
- Which organ showed the response?

You may have observed that the student could show the direction of source of sound if it came from left or right sides. The student failed to show correct directions if sound came either from front or back. We can see that our ears are located towards the left and the right side of our face. Now, if a source is located towards the left, sound produced from it would reach our left ear slightly before it reaches our right ear. This minute difference helps the brain to differentiate between directions correctly. In case of sound travelling from front or back, the source of sound is equidistant from our ears, thus, it becomes difficult to decide the direction when we are blind folded. If our eyes were open they would help in stating the direction correctly.

Do you know?



Dr. Shubha Tule is a leading scientist of our country who has made pioneering discoveries on the development of brain and identified genetic material that could have role in development of areas in the brain related to control of involuntary actions and learning. She has attained international acclaim. She was conferred the Shanti Swaroop Bhatnagar award in the year 2010 for her achievements. She is presently working at Tata Institute of Fundamental Research, Mumbai.

8.1.6 Exchange of information: Stimulus and response

Different types of informations both external and internal to our body keep influencing our body in many ways. Without a response at a proper instant these may affect our body adversely. Fox example when our hand touches a hot utensil it instantly moves away from it. The stimulus of heat passes to our hands through the cells that are in direct contact to the stimulus. They are then carried forward by nerves called as sensory nerves to the spinal cord and further to the brain. A response is effected against the stimulus from spinal cord or brain through nerves called as motor neurons. These influence the muscles of our arms in such a way that contraction and relaxation movements occur, eventually leading to the movement of our hand away from the hot utensil. The complete circuit from sensing a stimulus to effecting a response is called as a reflex action. See figure 7 which is a schematic representation of the process. Most reflex actions, that is, actions occurring involuntarily, are effected by the spinal cord.

- What would have happened in the given example if there was no response?
- If a utensil is hot you use it carefully, do you think your brain has any role in doing so?

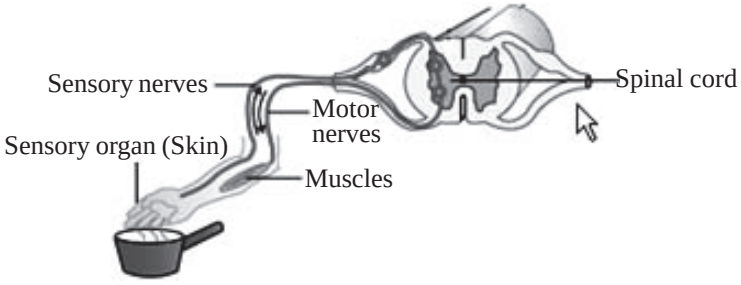


Figure-7: Reflex action- stimulus and response

Now if we consider what we had done in activity 1, we shall find that, we had been able to hold the stick quickly if we got intimation about its fall through our ear and we could observe the stick with our eyes. So, a response was effected by muscle movements against the stimulus to hold a falling stick. Thus, a coordination mainly between eye, ear and hand was necessary.

- Think and state which other parts of our body were involved in the action mentioned above.

If we try to represent the action of catching the scale schematically, it could be as in figure 8.

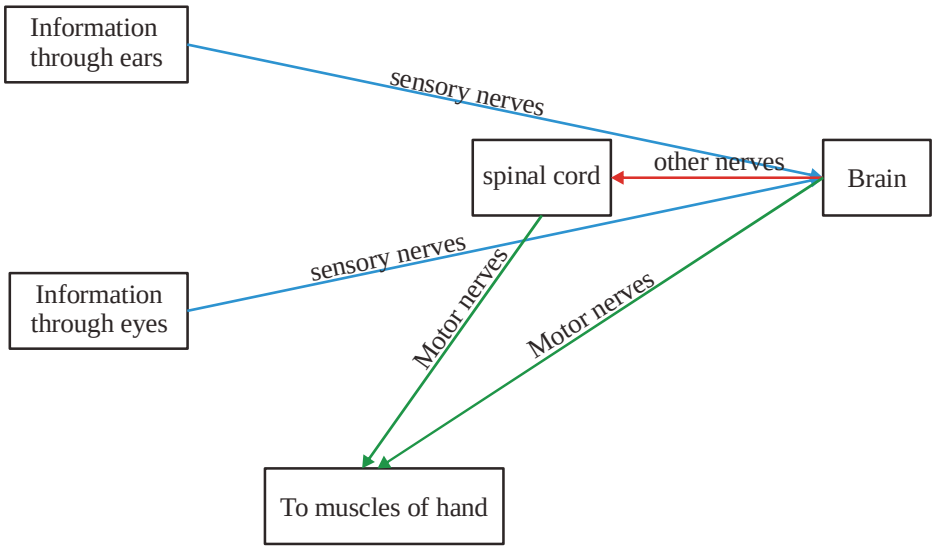


Figure-8: Schematic representation of pathway of information

- Write a few more examples of such reflex actions.
- What do you think happens when our fingers are pricked suddenly? What are the organs involved in the process? Draw a schematic diagram to represent the same.

Do you know?



Doctors often use a reflex hammer to strike the just below the knee joint to check for sensations. This produces a pressure signal which travels back to the spinal cord through the sensory nerves which synapse (without interneurons) directly with the motor nerves in the spinal cord. From there, the motor neuron conducts a response back to the muscles in thigh region, triggering contraction. This contraction, coordinated with the relaxation of the antagonistic muscles of lower portion of our leg causes the leg to kick. This is a reflex which helps maintain posture and balance, allowing tokeep one's balance with little effort or conscious thought. The pain felt later is due to the fact that the information of this act and the pain signals that were produced reach the brain from where responses are effected through a different pathway.

8.1.7 Conduction of information by hormones

Apart from the electrochemical pathways through a system of nerves there are other pathways of conduction of signals or informations in our body. There are certain organs where some chemicals are produced that reach different locations of our body through the blood. These chemicals are mainly proteins and called as 'hormones'(a word of Greek origin that means effecting motion).

Hormones are important agents that help in control and coordination between different parts of the body. Some examples of control and coordination by hormones are as follows.

Other than certain chemicals required for digestion, a hormone called insulin is also produced by our pancreas. Without the presence of insulin, our body fails to use sugar, thus the kidneys pass sugar present in blood into the urine. We are said to be affected by diabetes when we have sugar in urine and more sugar in blood than normal (a normal would a random measure between 80-200mg/ deciliter of blood sample). A very high amount of insulin is also fatal for the body. Say if a patient is given high dose of insulin, the kidneys remove all sugar available in blood leading to a dearth of sugar for cells even in our brain. Insulin is essential for each cell of our body. It is important that a proper amount of insulin reaches the cells, thus, its produced in a particular organ the pancreas. It is released and transported through blood to different parts of our body under the control of our nervous system.

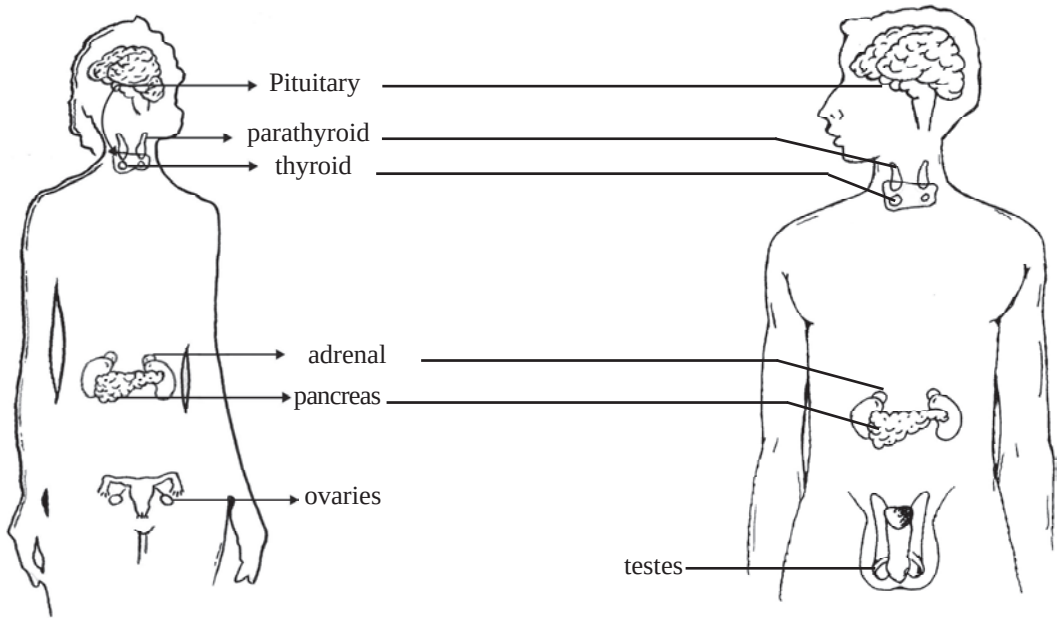


Figure-9: Some organs that produce hormones

There is a gland called 'Parathyroid' in our neck region that produces a hormone that is mainly responsible for maintaining level of Calcium in the blood. A low amount of the hormone leads to low levels of calcium in the blood leading to odd muscular cramps. Perhaps bones take up calcium from blood. A high level of hormone leads to high level of calcium in blood as the bones start softening as calcium passes into the blood from them. Both conditions are fatal for the body.

- Does nervous system have any role in the release of hormone from the parathyroid gland? What do you think?

Another gland called thyroid produces the hormone thyroxin which raises the rate of oxidation in our body. Without this hormone, rate of oxidation is negligible. If the level of this hormone is less in blood, a person becomes fat and sluggish. A high level of the hormone leads to loss of weight as the person becomes thin and hyperactive. The pulse rate increases and several other complications emerge and the person becomes seriously ill. When due to cancer or other complications, this organ is removed, the person is administered a fixed required amount of dosage of the hormone daily. If it is given at a higher dose than daily requirement then the oxidation rate escalates leading to high oxygen demand and the person becomes ill. If a normal required dosage was given to a person with a normally functioning thyroid, then the organ would stop producing the hormone as long as the level of the hormone administered externally remained normal in blood.

Adrenals present above are kidney produce several hormones. They are well known in producing hormones that effect actions related to responses during sudden fear or embarrassment. A particular hormone secreted by this gland helps in controlling level of salt in our body. This hormone is secreted from the outer layers of this gland. In the absence of production of this hormone a person may be kept alive by feeding her with a fixed small amount of salt per day otherwise it would be fatal for the individual.

Hormones are also produced by the testis in males and ovaries in females that bring about various changes during adolescence like development of beard, change of voice etc. in males and development of mammary glands, start of menstruation etc. in females. Hormone from these glands is released in mature individuals as well and helps in the production of reproductive cells eggs in females and sperms in males. Another hormone released in females called as 'progesterone' aids in pregnancy and menstruation.

8.1.8 Control of level of hormones

We have studied about the control of nervous system over secretion of hormones. Some of the glands themselves effect the secretion of other glands. Maximum varieties and amount of hormones are secreted by the pituitary gland (see fig 9 for its location). The hormones secreted by this gland have various roles in our body like growth and development, use of sugar, lactation in mothers' breast feeding their babies, etc. Apart from this, hormones secreted from pituitary control the release of hormones by several other glands like pancreas, thyroid, adrenal, testes ovary etc. The hormones released from other glands also affect the release of the respective pituitary hormone. For example, a hormone of pituitary gland controls the secretion of thyroxin from thyroid. If thyroxin level increases in blood the release of the pituitary hormone stops. The size of testes and ovaries increases during adolescence under the influence of pituitary hormones. These gonad stimulating hormones of pituitary glands also maintain levels of hormones secreted by ovaries and testes. A hormone related to growth, secreted by the pituitary acts antagonistically to the hormone insulin. The release of certain other hormones by the pituitary accelerates the release and activity of other hormones that may further influence the start and stop of the release of the respective pituitary hormone.

One such hormone is a type of adrenal gland stimulating hormone that stimulates the secretion of a hormone from the inner part of the adrenal glands. Our heart rate increases and blood flows into muscles at a higher rate during emotional situations or during exercise due to the effect of this adrenal hormone. The effect of this adrenal hormone as well as the stimulating hormone is very fast contrary to most hormones that slowly affect a function. The release of stimulating hormone stops when there is adequate amount of the required adrenal hormone in blood.

Progressive research in this field has established the fact that most of the actions in our body are carried out by a balance in two antagonistic pathways. We have come to know a lot about hormones now and if we do not use our knowledge properly, it may affect our health adversely.

A connection between nervous system and glands that release hormones exists all through our body, an important and extensive connection being formed in the brain with strands of nervous tissue and the pituitary. This connection between nervous system and glands secreting hormones aids in all life processes and proper functioning of our body. It is because of this connection that often your doctor suggests you to stay in high spirits during your illness so that you may be cured quickly.

8.2 Control and Coordination in plants



Figure-10: Touch-me-not plant

Research in the field of control and coordination in plants mainly started around 200 years back. Our knowledge acquired in these 200 years has helped us to believe that most life processes that occur in animals occur in a similar ways in plants. We have also realized that processes of control and coordination also occur in plants. You are aware of the shutting of leaves in the touch- me -not plant. There are several other functions like blooming of flowers in the morning in several plants while in some others like night queen blooming of flowers in the dark, closing of tamarind leaves at night, climbers climbing around a support, seeds sprouting and seedlings growing out of the soil are all examples of processes that indicate responses in plants to certain stimuli, light, temperature, presence of an obstruction etc. These show that there is conduction of information in some way.

Experiments started by Charles Darwin and his son Francis Darwin in the 1880's that there are certain chemicals that may be involved in the process of sensation in plants.

Scientists had been studying about the effect of various stimuli on plants. They found different types of responses usually effected by growth. Let us take an example. A certain study showed that if the tips of stems of plants were illuminated constantly from a certain direction, then cells in meristematic tissues in the opposite side of the stemstart dividing very fast leading to a bend in the stem towards the source of light. Thus it was speculated that some chemical passed on from tip to the region of growth bringing about the response. The presence of the chemical was later confirmed. It was isolated tested and named as 'Auxin'.

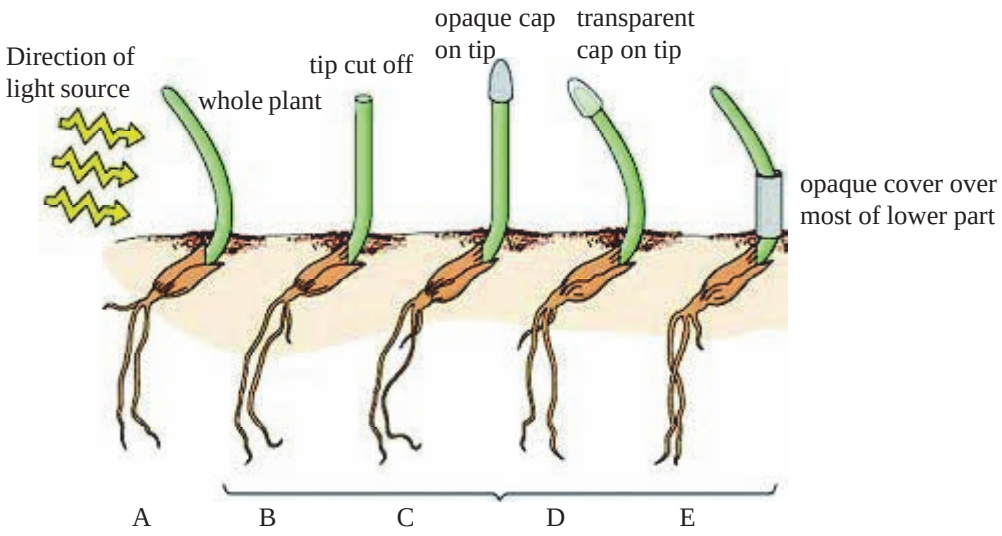


Figure-11: Experiment conducted by Charles and Francis Darwin (different areas of stem are covered for the same direction of light source in plants of similar size and same species).

- Observe the figure and state the conditions (A,B, C, D or E) in which the stem bends towards light. How do you think this happened?
- What inferences may be drawn from this experiment?

We now know that plant respond to various stimuli like presence of water, light, heat, touch, chemical substances etc. The informations of stimuli and the respective response is passed on in the form of certain chemicals in plants which many scientists call as 'plant hormones'. These chemicals aid in actions like opening and closing of stomata, falling of leaves, increase in rate of division of cells, growth of different parts, maturation of buds, blooming of flowers, ripening of fruits etc.

'Auxin' was found to be a growth hormone. Another growth hormone was found to be a different compound and named as 'gibberellin'. A chemical compound 'Cytokinin' is found in large amounts in seeds and fruits. It is found to increase the rate of cell division. Some chemicals like abscisic acid arrest growth in plants.

You may have observed that there are some plants around which no other plants are found to grow (eucalyptus being one of them, you may hardly see any undergrowth at the base of it). This is because a chemical is released from the roots of the plant that arrests the growth of other plants. When a ripe fruit is kept among unripe ones we often find the unripe fruits ripen quickly. This may be due to release of ethylene by some ripe fruits that influence the unripe ones.

You may have observed insects and birds flying around flowers and feeding on the nectar produced in them. This is a sugary fluid produced by glands that may be present on any part like sepal, petal, stamen, carpel etc. This fluid mainly helps to attract pollinators like the insects and birds bringing about pollination in plants. Another sugary fluid released over style attaches the pollen and induces its germination to form a tube carrying the reproductive cells through the style to the ovary. There are other chemicals as well that influence pollen germination.

All life processes whether they occur in a single cell or in multicellular bodies go on with the help of either chemicals or electrochemical means in controlled and coordinated manner.

Do you know?

Once in a certain area in the grasslands of South America, there was scarcity of water due to low rainfall. A type of tree similar to Babul/Acacia grows in those grasslands. The leaves of the plant contain tannin usually in very small amounts that does not have any effect on the animals that feed on it. During this period it was observed that deers and some other animals of the grassland started dying in large numbers. These animals could have moved out of the area due to scarcity of water but they did not. Moreover water scarcity did not seem to be the cause of their death. Scientists examining the dead animals found an alarming amount of tannin in their gut.

Studying the plants showed that the leaves also contained a very high amount of tannin, which was quite unlikely for the plants. Thus the scientists speculated that lowering of availability of water, had caused such a change that as the herbivores died out, these plants would have adequate water available for their survival. Apart from this it was found that trees even in areas where there wasn't any scarcity of water, the leaves still had high levels of tannin. The air had high amount of ethylene everywhere. It was found that the trees released ethylene in the air and it acted as a signal for other plants even in far of areas.

The source of information for this is available on https://www.youtube.com/watch?v=ZrXksBKRW1A

Key words

Control, coordination, sensory, motor, stimulus



What we have learnt

1. The nervous system and hormones in our body are mainly responsible for control and coordination.
2. The exchange of information in the body occurs by electrochemical or simply chemical means.
3. Reflex actions are controlled by the spinal cord. Brain does not have a major role. Information of the action is usually transmitted from spinal cord to the brain.
4. Chemical means of control are found in plants.
5. Hormones are chemical substances responsible for control and coordination.

Exercise

1. Choose the right option-

   (i) Which of the following is a plant hormone?

   (a) Insulin (b) Thyroxine

   (c) Estrogen (d) Cytokinin

   (ii) his is not a part of the structure of a nerve cell-

   (a) dendrite (b) nucleus

   (c) axon (d) cellulosic cell wall

   (iii) A gland present in our brain that controls hormonal secretions of several other glands in our body.

   (a) Pituitary (b) liver

   (c) adrenal (d) pineal

   (iv) A hormone from this gland controls the level of sugar in blood-

   (a) Parathyroid (b) Pancreas

   (c) Adrenal (d) Pineal

   (v) When we asleep at night and are bit by a mosquito, we often try to kill the mosquito. This action is controlled by-

   (a) Pituitary gland (b) spinal cord and brain

   (c) hormones (d) pineal gland

2. Make schematic line diagrams and describe a reflex action of our body.
3. Write a note on the role of spinal cord and brain on reflex action.
4. 'There is a coordination between the secretion of hormones and the nervous system', justify the statement.
5. Illustrate a nerve cell with a neat and labeled drawing and describe it.
6. How are involuntary actions controlled in our body, explain with a diagram?
7. Write an example of chemical control in plants.
8. Explain the need of control and coordination in any living organism with examples.
9. 'Plants are sensitive to light' cite an experiment to justify this statement.
10. Give examples to show how plants respond to stimuli.